MILITARY STANDARD

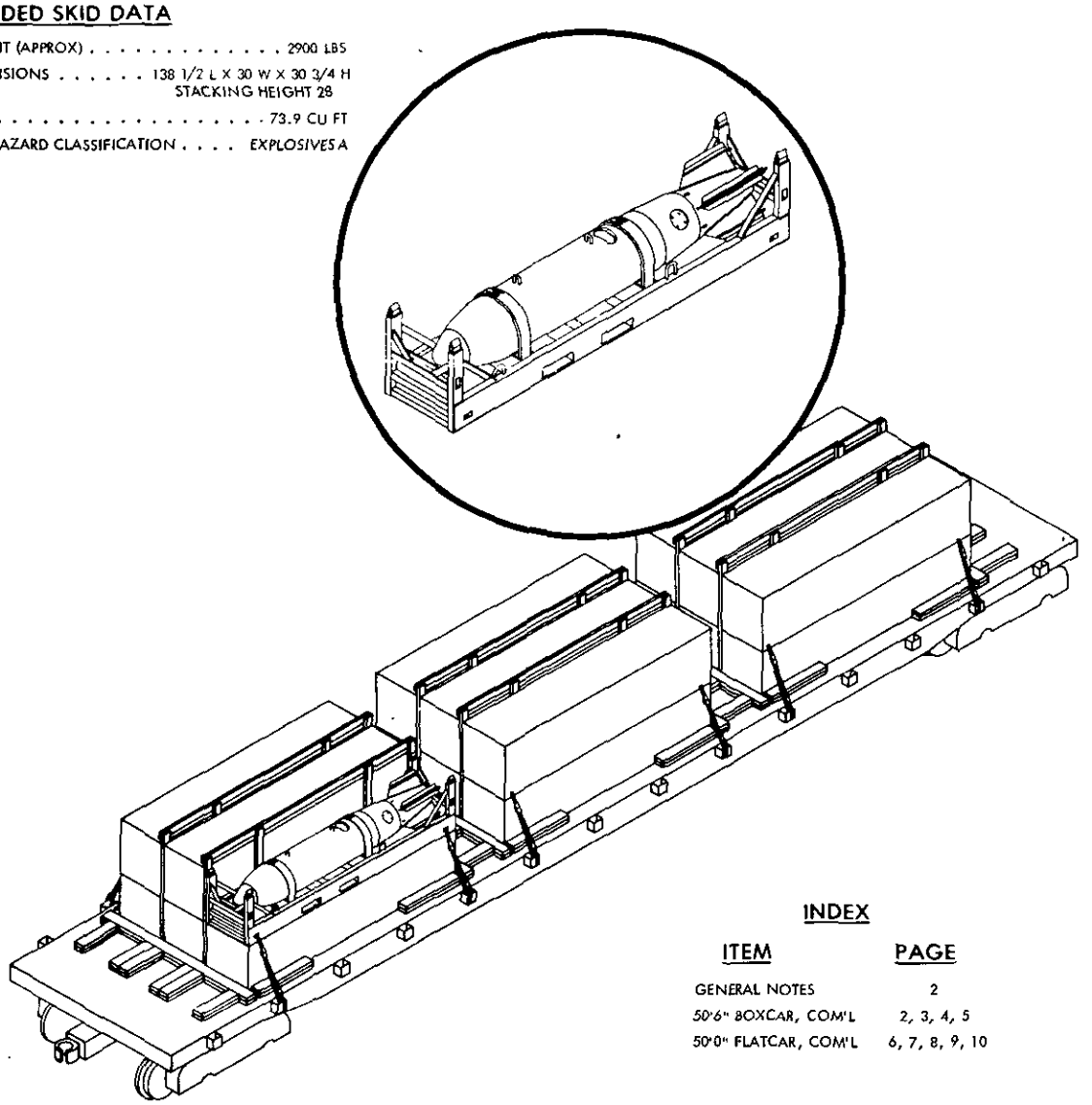
MIL-STD-1325-191
(NAVY)

20 MARCH 1981

RAILCAR LOADING
MINE, UNDERWATER MK 65 MOD 0 AND MODS IN SKID MK 25 MOD 2

LOADED SKID DATA

WEIGHT (APPROX) . . . . . 2900 LBS
DIMENSIONS . . . . . 138 1/2 L X 30 W X 30 3/4 H
STACKING HEIGHT 28
CUBE . . . . . 73.9 CU FT
DOT HAZARD CLASSIFICATION . . . . EXPLOSIVES A



INDEX

ITEM	PAGE
GENERAL NOTES	2
50'6" BOXCAR, COM'L	2, 3, 4, 5
50'0" FLATCAR, COM'L	6, 7, 8, 9, 10

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

TECHNICAL DIRECTION AGENT (TDA) DATE

BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

SIGNATURE SUPERVISOR, MILITARY AND INTERMODEL SERVICES DATE

ORIGINATOR DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT.
3. LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH WITH STAGGERED 15 FT DOORS, AND COMMERCIAL FLATCARS WITH A PLATFORM 50 FT 0 INCHES LONG AND 9 FT 0 INCHES WIDE WITH STAKE POCKETS SPACED 42 INCHES ON CENTERS. FLATCARS WITH STEEL FLOOR ENDS AND/OR STEEL BOLSTERS SHALL NOT BE USED.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF MK 65 MOD 0 AND MODS UNDERWATER MINES IN MK 25 MOD 1 SKIDS.
7. THE SKIDS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK AND/OR MK 45 MOD 0 HANDLIFT TRUCKS.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:

    DUNNAGE LUMBER - FED. SPEC MM-L-751

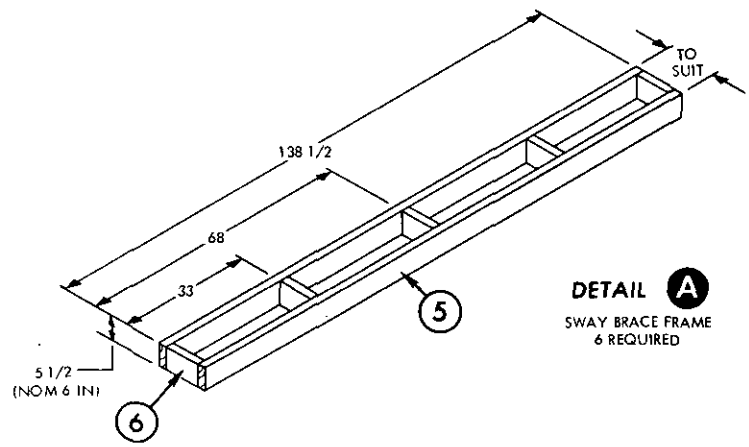
    NAILS - FED. SPEC FF-N-105

    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, FINISH A, B, OR C

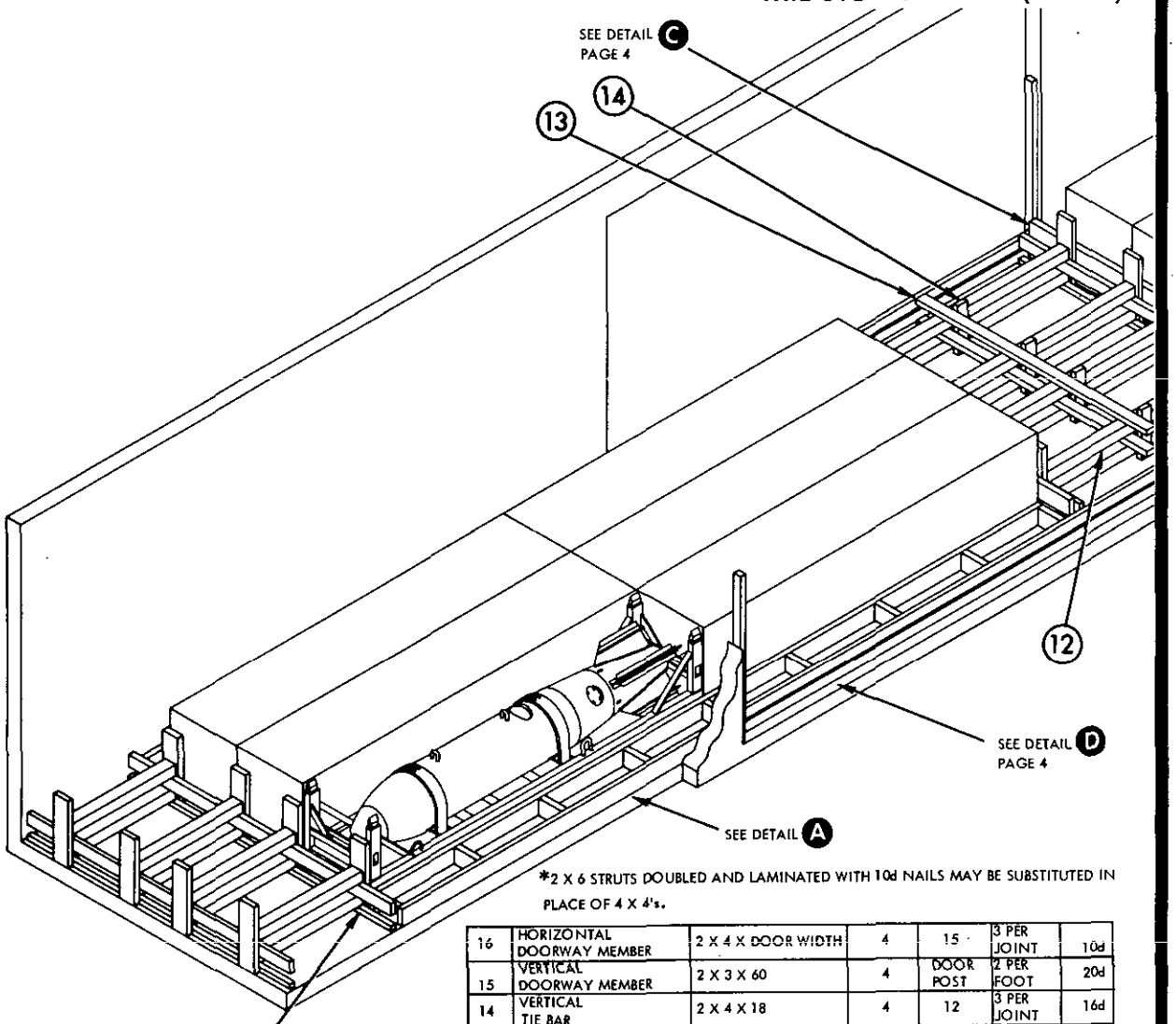
    SEALS - FED. SPEC QQ-S-781, STYLE II, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1).

50 FT 6 IN. BOX CAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 9 MINES IN SKIDS (1 LAYER HIGH) WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS WITH WOOD LINING, DUMMY SKIDS (SEE DETAIL E, PAGE 5) MAY BE USED IN PLACE OF THE MISSING LOADED SKIDS AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) MAY BE USED. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. IN ALL STEEL BOXCARS DUMMY SKIDS MAY BE USED IN PLACE OF LOADED SKIDS. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 15 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE NOT APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS LESS THAN 15 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS. THE PLUG DOORS MUST BE SPANNED WITH A 2 X 6 X DOORWIDTH + 24 INCH PIECE NAILED TO THE SIDE WALLS (5-10d NAILS EACH END) AT THE FLOOR LEVEL. THE WIDTH AND LENGTH OF THE SWAY BRACES IN THE DOORWAY AREA WILL HAVE TO BE ADJUSTED AS NECESSARY. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. DIMENSIONAL LUMBER DOORWAY PROTECTION, PIECES 17 AND 18, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED EXCEPT WHEN CAR HAS COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DIMENSIONAL LUMBER DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. PLUG DOORS MUST BE SECURELY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



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*2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4's.

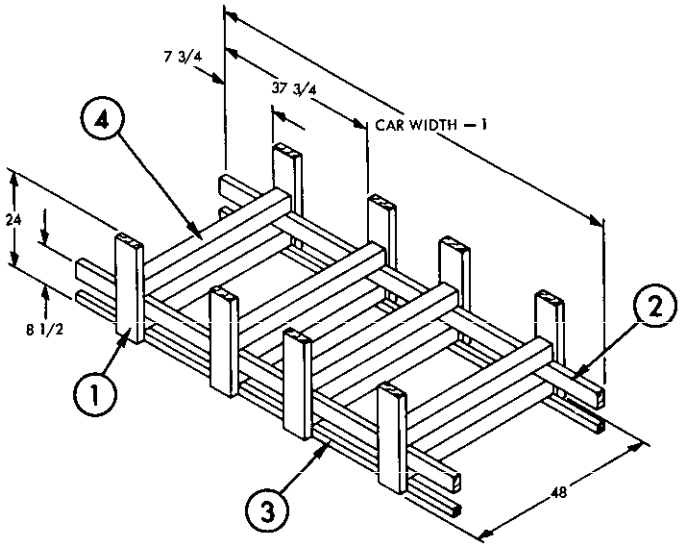
PC. NO.	DESCRIPTION	SIZE	NO. PCS REQ'D.	NAIL TO	NUMBER NAILS	SIZE
16	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	4	15	3 PER JOINT	10d
15	VERTICAL DOORWAY MEMBER	2 X 3 X 60	4	DOOR POST	2 PER FOOT	20d
14	VERTICAL TIE BAR	2 X 4 X 18	4	12	3 PER JOINT	16d
13	HORIZONTAL TIE BAR	2 X 4 X 96	2	12	3 PER JOINT	16d
12	STRUT	4 X 4 X WEDGE FIT*	8	8	2 PER JOINT	10d
11	CENTER GATE HOLD DOWN	2 X 4 X CUT TO FIT	4	7	4	10d
10	CENTER GATE STRUT CLEAT	2 X 2 X CAR WIDTH-1	2	8	2 PER JOINT	10d
9	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	2	8	3 PER JOINT	10d
8	CENTER GATE VERTICAL	2 X 6 X 24	8	SEE 7	—	—
7	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	4	8	3 PER JOINT	10d
6	SWAY BRACE CROSS PIECE	2 X 6 X CUT TO FIT	30	SEE 5	—	—
5	SWAY BRACE LONGITUDINAL	2 X 6 X 138 1/2	12	6	3 PER JOINT	16d
4	END GATE STRUT	4 X 4 X 45*	16	1	2 PER JOINT	16d
3	END GATE STRUT CLEAT	2 X 2 X CAR WIDTH-1	4	1	2 PER JOINT	10d
2	END GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	4	1	3 PER JOINT	10d
1	END GATE VERTICAL	2 X 6 X 24	16	SEE 2,3	—	—

LIST OF MATERIALS AND NAILING DATA

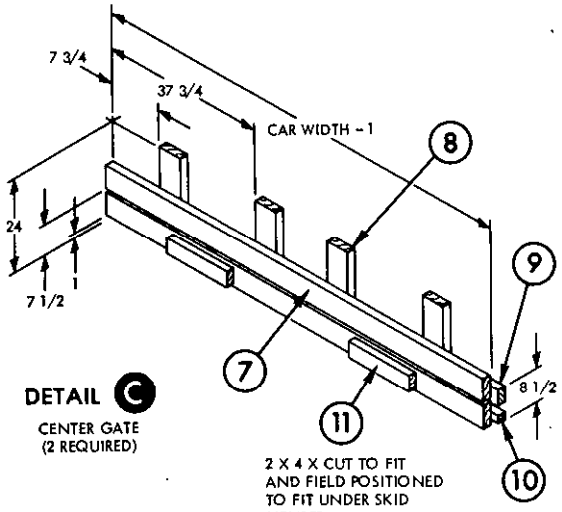
CARLOAD DATA

NUMBER OF LOADED SKIDS 9
LOAD WEIGHT (APPROX) . . . . . 26,100 LBS
DUNNAGE WEIGHT (APPROX) . . . . . 1,200 LBS
CARLOAD WEIGHT (APPROX) . . . . . 27,300 LBS

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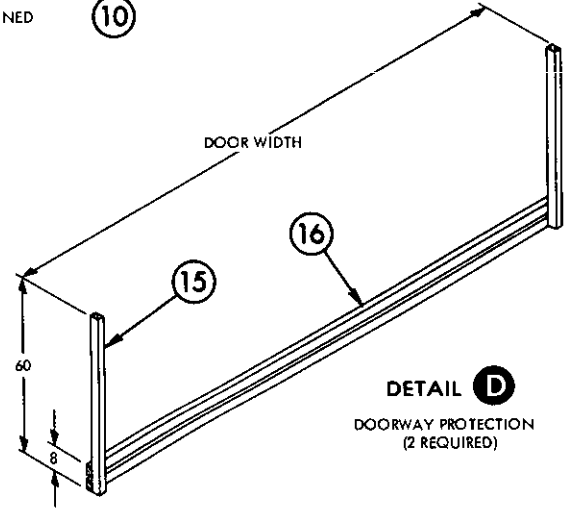


DETAIL B
END GATE ASSEMBLY
(2 REQUIRED)



DETAIL C
CENTER GATE
(2 REQUIRED)

2 X 4 X CUT TO FIT
AND FIELD POSITIONED
TO FIT UNDER SKID
MEMBER

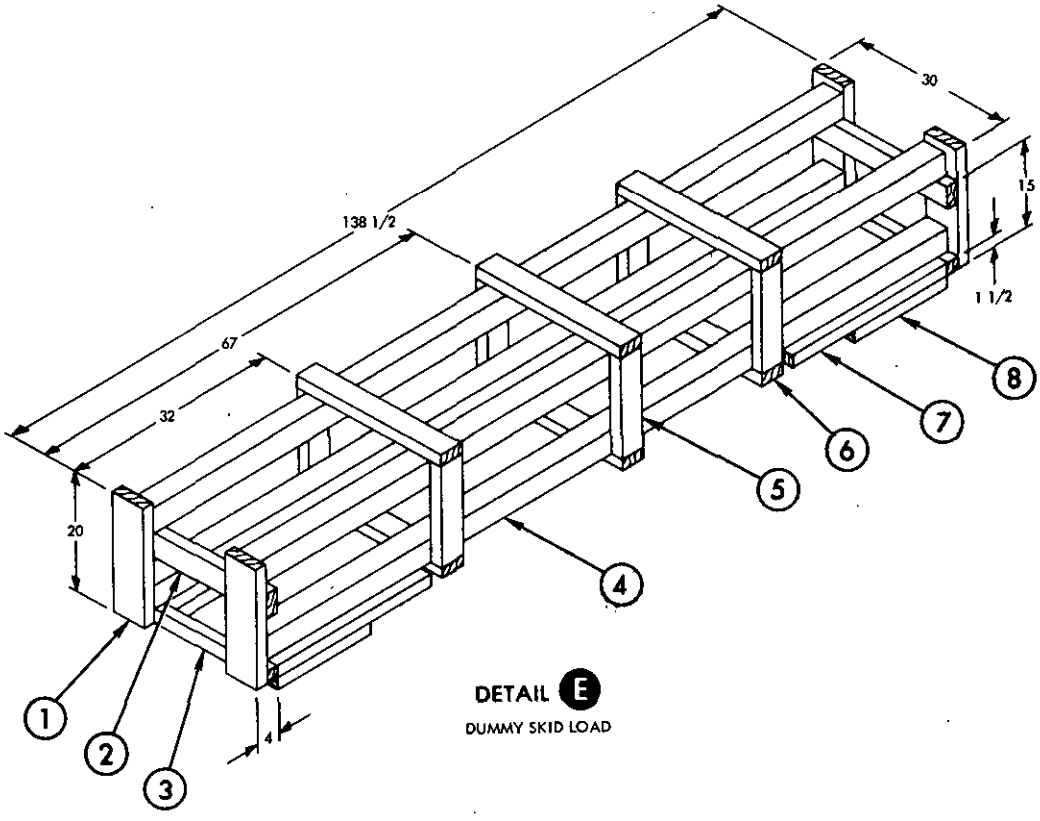


DETAIL D
DOORWAY PROTECTION
(2 REQUIRED)

WHEN BOXCAR HAS STEEL DOOR POSTS,
DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY)
IS REQUIRED.

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DUMMY SKID LOAD



DETAIL E
DUMMY SKID LOAD

(1) TOE NAIL STRUT TO VERTICAL WITH 16d NAILS, 3 EACH END (TOP AND BOTH SIDES) ALSO END NAIL THROUGH VERTICAL WITH 16d NAILS, 2 EACH END.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
8	STOP	2 X 6 X 18	4	7	7	16d
7	SPACER	2 X 6 X 30	4	4	4	16d
6	HORIZONTAL SUPPORT PIECE	2 X 4 X 30	6	4	3 PER JOINT	16d
5	VERTICAL SUPPORT PIECE	2 X 4 X 17	6	4	3 PER JOINT	16d
4	STRUT	4 X 4 X 138 1/2	4	1	SEE NOTE 1	
3	STRUT CLEAT	2 X 2 X 30	2	1	2 PER JOINT	16d CLNHD
2	STRUT CLEAT	2 X 4 X 30	2	1	3 PER JOINT	16d CLNHD
1	VERTICAL	2 X 6 X 20	4	SEE 2,3	—	—

LIST OF MATERIALS & NAILING DATA

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50 FT. 0 IN. FLATCAR, COMMERCIAL

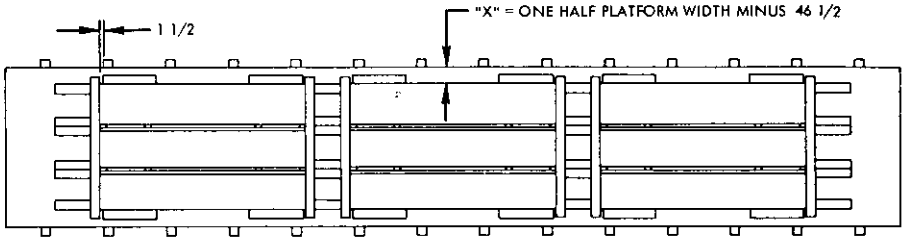
THE CARLOAD CONSISTS OF 18 SKIDS WHICH MUST BE DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED ON COMMERCIAL FLATCARS, THE PROCEDURE OUTLINED ON PAGE 11 SHALL BE FOLLOWED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH (SEE MIL-STD-1325 (NAVY).

SKIDDED LOADS STACKED TWO HIGH SHALL BE STRAPPED TOGETHER WITH FOUR BANDS OF STEEL STRAPPING, (PIECE 1), POSITIONED AS SHOWN (SEE END VIEW, PAGE 8) AND SECURE WITH TWO DOUBLE CRIMPED SEALS (PIECE 2) PER STRAP. AS ROWS ARE PLACED AGAINST EACH OTHER, PLACE AN ANTI-CHAFING FRAME, (SEE DETAIL A), BETWEEN EACH SKIDDED ROW.

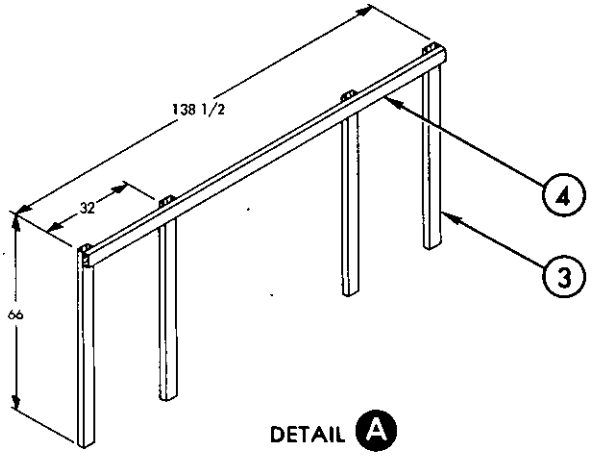
STRAP THE TOP LAYER OF EACH STACK OF SKIDDED LOADS TOGETHER WITH TWO BANDS OF STEEL STRAPPING EACH END, (PIECES 13 & 14) POSITIONED AS SHOWN (SEE END VIEW, PAGE 8), AND SECURE WITH TWO DOUBLE CRIMPED SEALS (PIECE 2) PER STRAP.

STRAP THE LOAD TO THE CAR WITH TWO BANDS OF STEEL TIE DOWN STRAPPING OVER EACH STACK OF SIX SKIDDED LOADS, POSITIONED AS SHOWN. TENSION AND SECURE THE TIE DOWN STRAPPING (PIECE 15) TO THE CAR AS FOLLOWS:

A. RUN STRAPPING FROM SIDE OF THE LOAD DOWN TO AND UP THROUGH THE STAKE POCKET.
B. POSITION STAKE POCKET PAD (PIECE 16) IN PLACE ON ONE SIDE OF THE CAR AND SECURE WITH SEAL (PIECE 18).
C. CONTINUE STRAPPING (PIECE 15) BACK UP AND THROUGH THE STRAPPING SLOT OF THE UPPER LAYER SKID FRAMES.
D. POSITION EDGE PROTECTOR PADS (PIECE 17) IN PLACE OVER THE OUTBOARD EDGES OF THE OUTER ROW SKID FRAME SLOT AND SECURE WITH SEALS (PIECE 18).
E. CONTINUE STRAPPING DOWN TO AND THROUGH THE STAKE POCKET ON THE OPPOSITE SIDE OF THE CAR.
F. POSITION STAKE POCKET PAD AND SECURE WITH SEAL.
G. POSITION STRAPPING BOARD (PIECE 19) OVER STRAP AS SHOWN AND RUN TIE DOWN STRAPPING BACK UP AND OVER THE STRAPPING BOARD AND DOWN THE SIDE OF THE LOAD WHERE IT IS TENSIONED AND SEALED WITH TWO DOUBLED CRIMPED SEALS.



LOADING PLAN



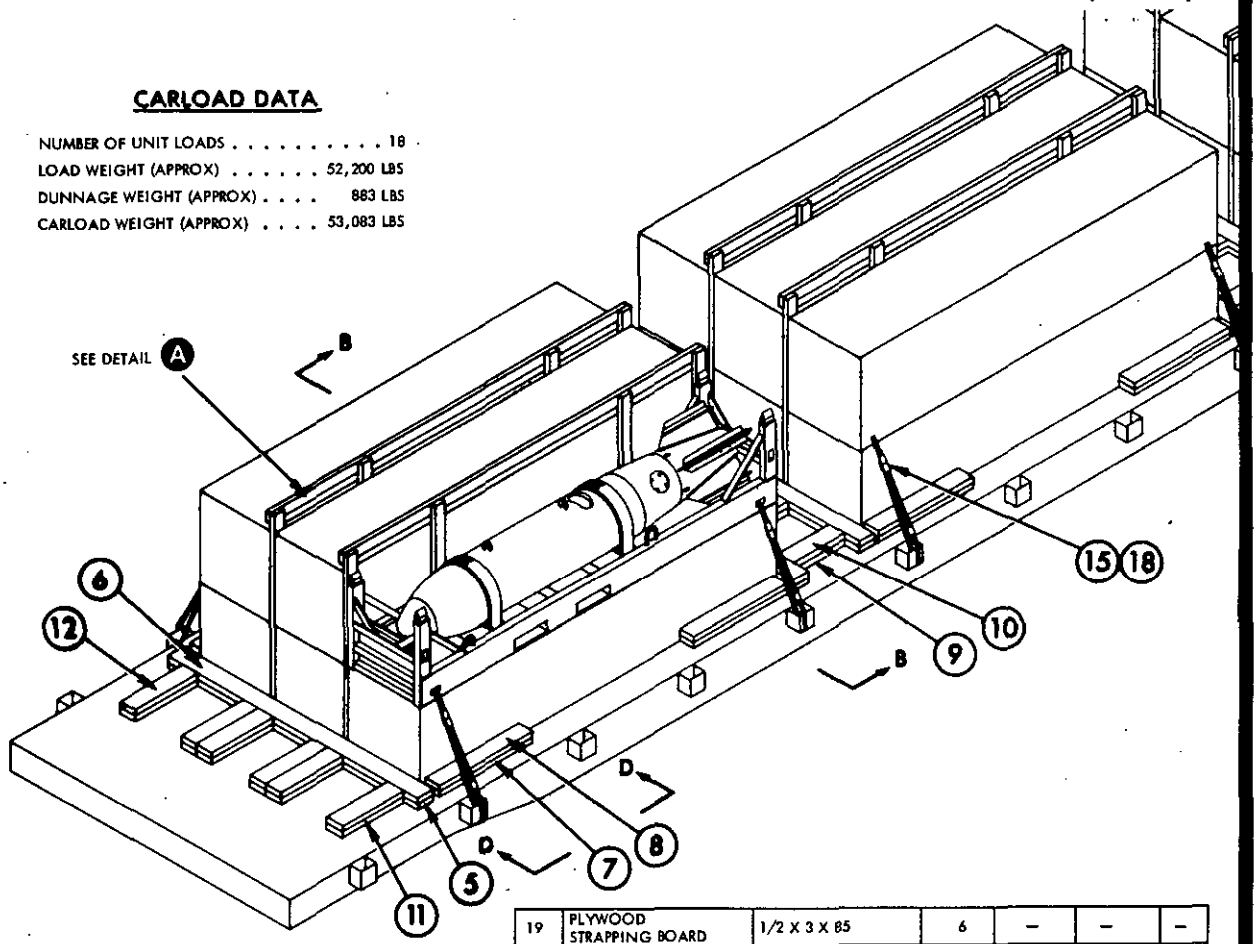
DETAIL A

ANTI-CHAFING ASSEMBLY
6 REQUIRED

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CARLOAD DATA

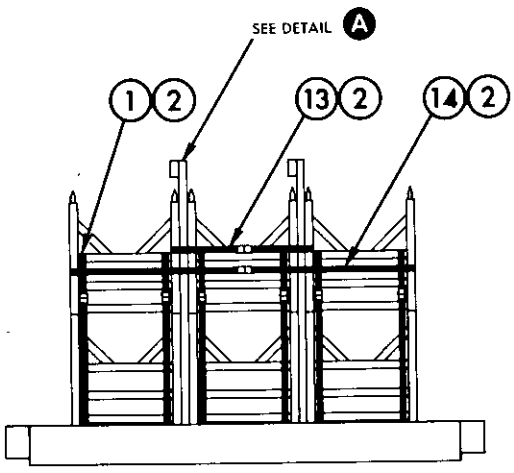
NUMBER OF UNIT LOADS 18
LOAD WEIGHT (APPROX) . . . . . 52,200 LBS
DUNNAGE WEIGHT (APPROX) . . . . 883 LBS
CARLOAD WEIGHT (APPROX) . . . . 53,083 LBS



PC. NO.	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO TO	NUMBER NAILS	SIZE NAILS
19	PLYWOOD STRAPPING BOARD	1/2 X 3 X 85	6	—	—	—
18	SEAL	2	36	—	—	—
17	EDGE PROTECTOR PAD	2 X .050 X 18	12	—	—	—
16	STAKE POCKET PAD	2 X .050 X 24	12	—	—	—
15	TIE DOWN STRAPPING	2 X .050 X 24 FT	6	—	—	—
14	CROSS STRAPPING	1 1/4 X .035 X 17 1/2 FT	6	—	—	—
13	CROSS STRAPPING	1 1/4 X .035 X 8 1/2 FT	6	—	—	—
12	END BLOCKING	2 X 6 X 24	12	11	4	60d
11	END BLOCKING	2 X 6 X 24	12	CAR FLOOR	4	30d
10	INTERMEDIATE BLOCKING	2 X 6 X 19 1/2	12	9	3	60d
9	INTERMEDIATE BLOCKING	2 X 6 X 19 1/2	12	CAR FLOOR	3	30d
8	SIDE BLOCKING	2 X 6 X 36	12	7	6	30d
7	SIDE BLOCKING	2 X 6 X 36	12	CAR FLOOR	6	30d
6	CROSS MEMBER	2 X 6 X 96	6	5	1 EVERY 8 INCHES	60d
5	CROSS MEMBER	2 X 6 X 96	6	CAR FLOOR	1 EVERY 8 INCHES	30d
4	ANTI-CHAFING TIE PIECE	2 X 4 X 11 FT 6 1/2	6	3	2 PER JOINT	10d
3	ANTI-CHAFING VERTICAL	2 X 4 X 66	24	SEE 4	—	—
2	SEAL	1 1/4	96	—	—	—
1	VERTICAL STRAPPING	1 1/4 X .035 X 9 FT	36	—	—	—

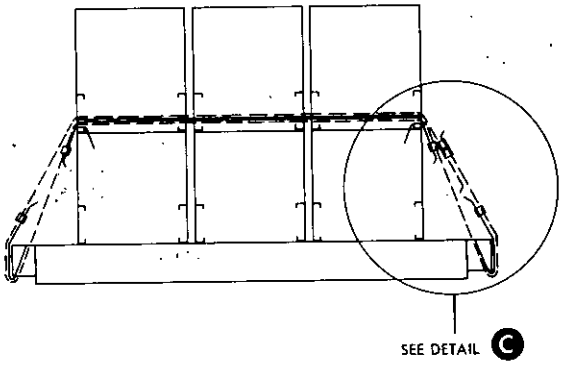
LIST OF MATERIALS AND NAILING DATA

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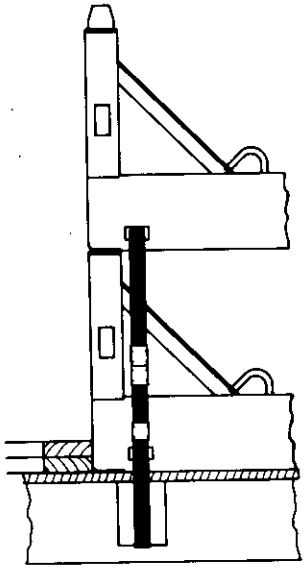
END VIEW

SHOWING VERTICAL STRAPPING
AND CROSS STRAPPING
(TIE DOWN STRAPPING AND FLOOR
BLOCKING OMITTED FOR CLARITY)



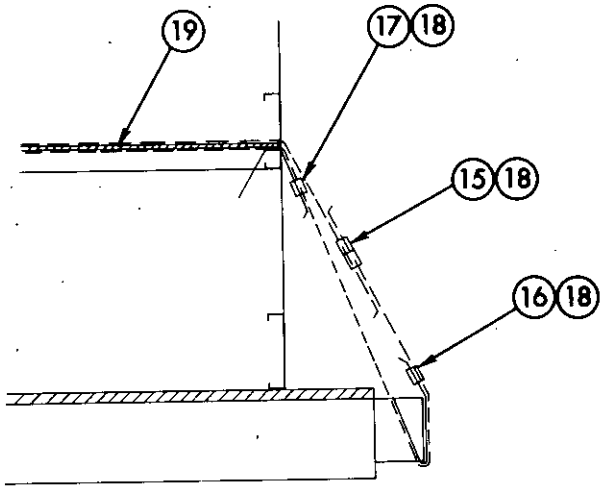
SECTION B-B

SHOWING DOUBLE THREADED TIE DOWN
STRAPPING WITH EDGE PROTECTOR PAD, STRAPPING
BOARD, AND STAKE POCKET PAD



VIEW D-D

SHOWING TIE DOWN STRAPPING SLOT
LINED UP WITH STAKE POCKET



DETAIL C

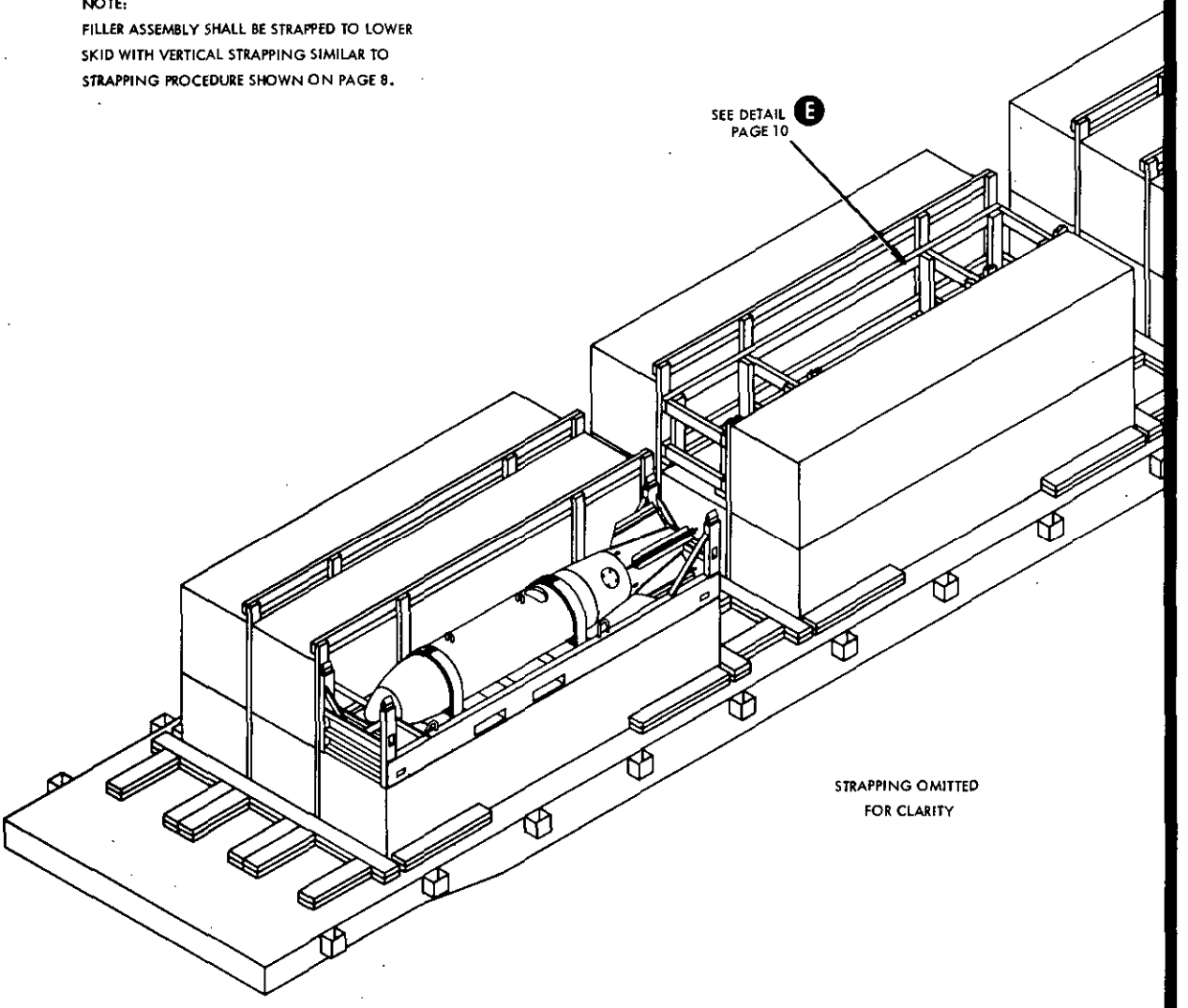
SHOWING STAKE POCKET PAD
EDGE PROTECTOR PAD, AND
PLYWOOD STRAPPING BOARD

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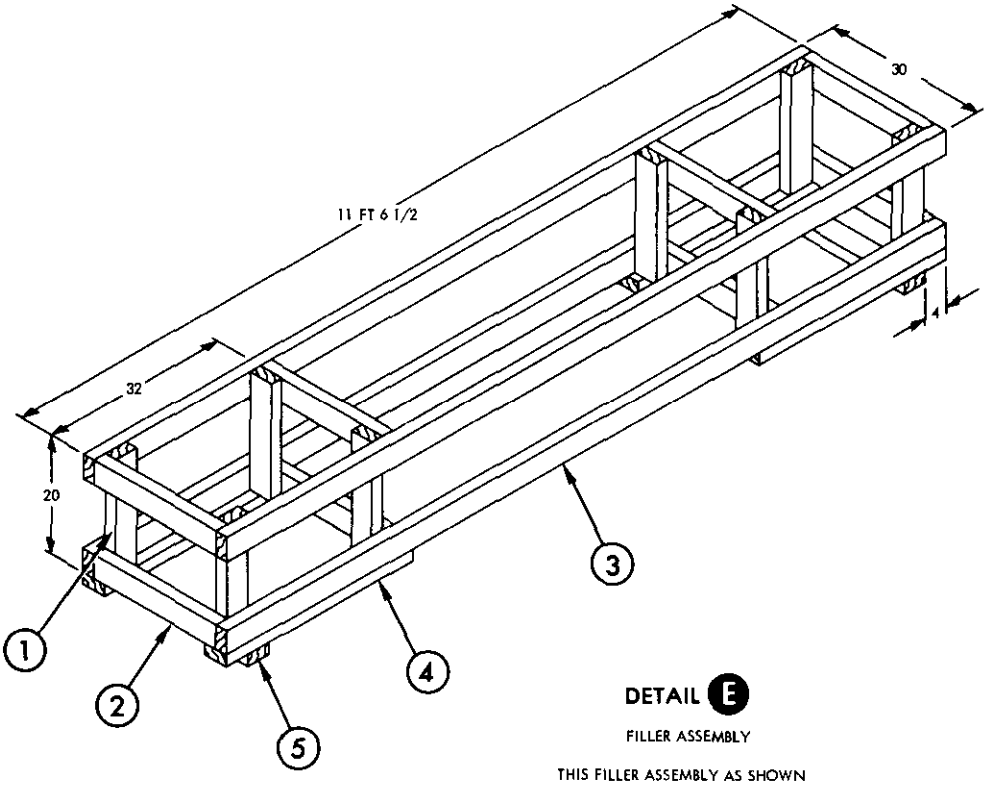
LESS THAN CARLOAD

USE FILLER ASSEMBLY (DETAIL E) AS SHOWN WHEN SHIPPING LCL. POSITION FILLER ASSEMBLY TOP LAYER, CENTER ROW OF ONE OR TWO STACKS AS REQUIRED.

NOTE:
FILLER ASSEMBLY SHALL BE STRAPPED TO LOWER SKID WITH VERTICAL STRAPPING SIMILAR TO STRAPPING PROCEDURE SHOWN ON PAGE 8.



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DETAIL E

FILLER ASSEMBLY

THIS FILLER ASSEMBLY AS SHOWN IS TO BE USED WHEN REQUIRED IN FLATCAR "LESS THAN CARLOADS" IN PLACE OF AN OMITTED SKID.

*QUANTITIES SHOWN ARE FOR ONE FILLER ASSEMPLY

PIECE NO.	DESCRIPTION	SIZE	NO. *OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
5	STOP	2 X 4 X 30	2	4	3 PER JOINT	10d
4	SUPPORT	2 X 4 X 36	4	2	2 PER JOINT	10d
3	LONGITUDINAL	2 X 4 X 11 FT 6 1/2	4	2	1 PER JOINT	10d
2	HORIZONTAL	2 X 4 X 27	8	1	3 PER JOINT	10d
1	VERTICAL	2 X 4 X 20	8	3	2 PER JOINT	10d

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